

Product and Category Management System

Task Requirements:

1. Design tables for Product and Category entities.
2. Define primary and foreign key relationships.
3. Store product details such as name, category, price, and stock.
4. Perform product insertion, updating, and deletion.
5. Generate category-wise product reports.

SQL Code

Create Category Table

```
CREATE TABLE Category (  
CategoryID INT PRIMARY KEY,  
CategoryName VARCHAR(50) NOT NULL  
);
```

Create Product Table

```
CREATE TABLE Product (  
ProductID INT PRIMARY KEY,  
ProductName VARCHAR(100) NOT NULL,  
CategoryID INT,  
Price DECIMAL(10,2),  
Stock INT,  
FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)  
);
```

Insert Categories

```
INSERT INTO Category VALUES  
(1,'Electronics'),  
(2,'Clothing'),  
(3,'Books');
```

Insert Products

```
INSERT INTO Product VALUES  
(101,'Laptop',1,55000.00,10),  
(102,'Smartphone',1,25000.00,15),  
(103,'T-Shirt',2,800.00,50),  
(104,'Database Book',3,650.00,20);
```

Update

```
UPDATE Product SET Price=60000.00, Stock=8 WHERE ProductID=101;
```

Delete

```
DELETE FROM Product WHERE ProductID=104;
```

Category-wise Report

```
SELECT c.CategoryName,p.ProductName,p.Price,p.Stock  
FROM Category c JOIN Product p  
ON c.CategoryID=p.CategoryID  
ORDER BY c.CategoryName;
```